

VAIBHAV PANCHAL

Junior AI/ML Developer | AI & Deep Learning Researcher | Full-Stack Engineer

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PROFESSIONAL SUMMARY

Junior AI/ML Developer at TestGrid Labs Inc. and I completed my Master's in Software Engineering at L.D. College of Engineering. Specializes in Machine Learning, Deep Learning, Computer Vision, and AI-driven automation. Research experience in oriented object detection with YOLOv11 + Efficient Channel Attention. Proven track record across 10+ projects spanning AI-powered platforms, full-stack web applications, and intelligent testing tools. Passionate about solving real-world problems through AI research and engineering.

WORK EXPERIENCE

Junior AI/ML Developer | **TestGrid Labs Inc.** | Navsari, Gujarat (on-site) *May 2026 – Present*

- Building AI-driven solutions and contributing to innovative automated testing technologies.
- Developing intelligent pipelines leveraging ML/DL for test automation and quality assurance at scale.
- Collaborating with cross-functional teams on AI research and modern software engineering practices.

Trainee Associate Software Developer | **Techreale Pvt. Ltd.** | Valsad, Gujarat *Jan 2024 – Apr 2024*

- Developed and debugged features for the Famyll healthcare platform (React.js, MUI, Node.js, MongoDB).
- Integrated RESTful APIs and built responsive, user-centric interfaces; collaborated via Git & GitHub.
- Participated in Agile sprints; contributed to claim management, auth flows, and real-time communication modules.

React Developer Intern | **INFOLABZ IT Services Pvt. Ltd.** | Ahmedabad, Gujarat *Jul 2023 – Aug 2023*

- Built responsive React.js UIs with effective state management and async API integration.
- Debugged and optimized component performance; strengthened skills in JavaScript and REST APIs.

PROJECTS

Movie Recommendation System

Jan – Feb 2025

Stack: Python, Pandas, Scikit-learn, TF-IDF, Cosine Similarity, Streamlit

- Content-based recommender on TMDB 5000 dataset using TF-IDF vectorization and cosine similarity.
- Streamlit front-end for real-time recommendations; focused on scalability and clean UI.

Intrusion Detection System (ML + Deep Learning)

Apr – May 2025

Stack: Python, TensorFlow/Keras, Scikit-learn, NumPy, Pandas

- Hybrid IDS classifying network traffic as benign or attack using CNN + BiLSTM + MultiHead Attention.
- Achieved 98.4%+ accuracy; applied PCA and AutoEncoder for dimensionality reduction and feature compression.
- Benchmarked against Random Forest and SVM; evaluated with precision, recall, and F1-score metrics.

ExamForge – AI Exam Question Generator

2026

Stack: Python, Streamlit, Ollama, LLMs, OpenAI

- AI-powered platform that converts syllabus PDFs into structured question papers with configurable difficulty levels.
- Solves educator pain point of manually drafting curriculum-aligned test questions using LLM pipelines.
- GitHub: github.com/Rathi-patel/examforge

RESEARCH & PUBLICATIONS

Channel Attention for Oriented Object Detection in Remote Sensing | 2025

- Integrated Efficient Channel Attention (ECA) module into YOLOv11s-OBb for improved remote sensing detection.
- Evaluated on the DIOR-R dataset; achieved +3.8% mAP50 improvement and +4.3% recall gain.
- Enhanced localization of small/dense objects while maintaining lightweight model efficiency.
- Domain: Computer Vision, Remote Sensing, Attention Mechanisms, Object Detection.

TECHNICAL SKILLS

Programming	Python (95%), SQL, Java, C, C++, JavaScript, PHP, HTML, CSS
AI / ML	TensorFlow, Keras, PyTorch, Scikit-learn, Transformers, FastAPI, Flask
CV & Research	YOLOv11, ECA Attention, Object Detection, Remote Sensing, LLMs, RAG
Web & Tools	ReactJS, Node.js, MongoDB, Firebase, Docker, Git, Jupyter, Postman, Streamlit
Automation / Testing	Selenium Grid, Cross-browser Testing, FastAPI

EDUCATION

Examination	University	Institute	Year	CGPA / %
M.E. – Software Engg.	GTU (SE)	LDCE Ahmedabad	2024 – 2026	7.95
B.E. – Computer Engg.	GTU (CS)	GDEC Navsari	2020 – 2024	8.11
HSC	GSEB	AB School	2020	72%
SSC	GSEB	AB School	2018	69%

ACHIEVEMENTS

- SSIP Hackathon 2022 – Built a smart e-Hajiri-Master automated attendance management system (PHP).
- Certificate of Participation – Dev-o-lution Hackathon.
- 10+ projects completed spanning AI/ML, full-stack web, automation, and research domains.
- 2 research experiments / papers in CV and Cybersecurity (Deep Learning).

INTERESTS & HOBBIES

Technical: AI/ML, LLM & RAG, Computer Vision, Deep Learning, Automation, Cloud Computing, Data Science

Hobbies: Coding, Chess, Exploring new technologies, Gaming, Music